Data Appendix

Early Warning Signals in Federal Prisons: A Time-Series Analysis of Inmate Complaints
Snow Stormers | DS 4002 | Spring 2026

Section 1: Federal Inmate Complaint Filings (BOP SENTRY)

Each row of this dataset represents one complaint filing submitted by a federal inmate through the Bureau of Prisons' Administrative Remedy Program. A single grievance case may generate multiple filings if the inmate escalates through the three appeal levels: Administrative (Warden), Regional (Director), and Final (Central Office). The dataset was obtained via a Freedom of Information Act request by the Data Liberation Project in 2024 and covers filings received between January 2000 and December 2023 across 220 federal facilities.

Descriptive Statistics - Monthly Complaint Counts:

Statistics	Value
Total filings (2000-2023)	1,760,608
Months covered	288 (Jan 2000 - Dec 2023)
Unique facilities	220
Monthly mean	6,113
Monthly median	6,020
Monthly std. deviation	1,311
Monthly minimum	3,160 (Nov 2001)
Monthly maximum	12,240 (June 2020)

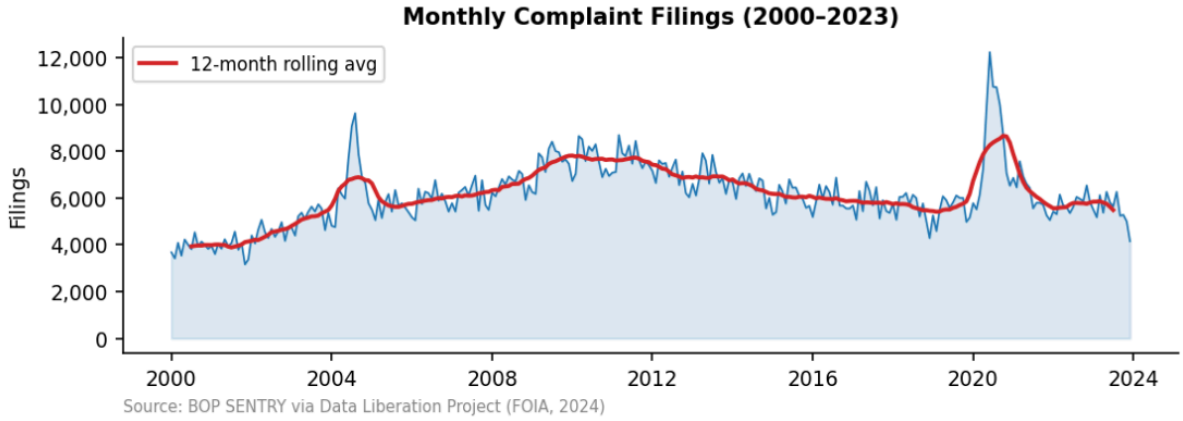


Figure 1 — Monthly complaint filings with 12-month rolling average. The series shows a wave structure peaking around 2011, declining through 2019, and spiking sharply in 2020.

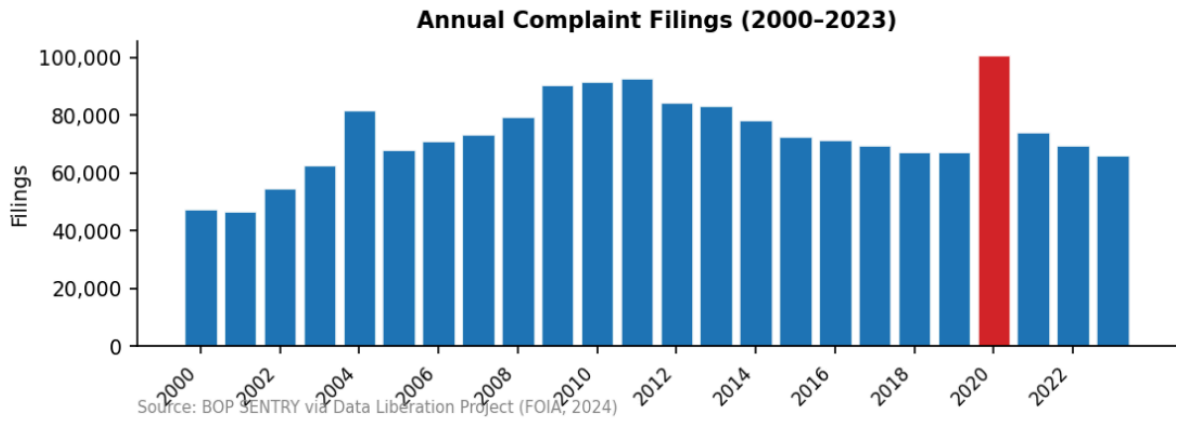


Figure 2 — Annual complaint totals. The 2020 bar (red) represents a 50% year-over-year increase driven by COVID-19 pandemic conditions inside federal facilities.

Appeal Level Distribution

Over the full study period, 44.1% of filings (776,516) were submitted at the Final (Central Office) level, 39.1% (688,938) at the Regional level, and 16.8% (295,154) at the Administrative (Warden) level. The predominance of Final-level filings reflects the multi-step escalation structure of the Administrative Remedy Program, in which many complaints are re-filed at successively higher levels after denial.

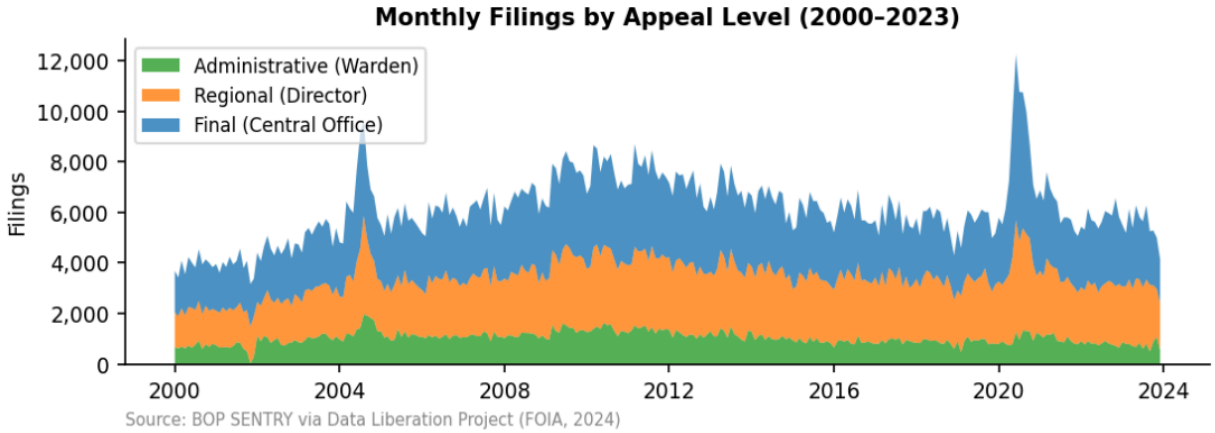
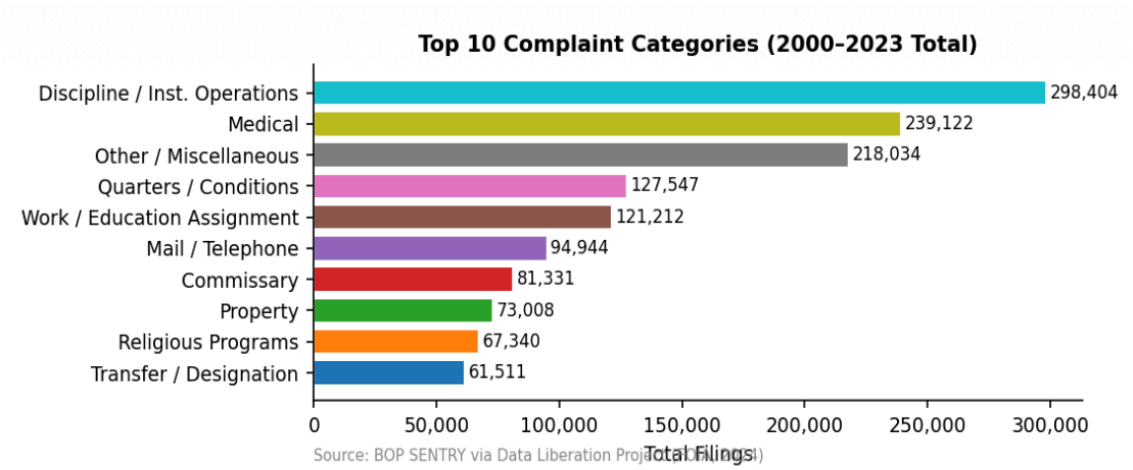


Figure 3 — Monthly filings by appeal level. The relative proportions of each level are stable across the full 24-year period.

Complaint Subject Categories

Each filing is assigned a primary subject code (field CDSUB1PR) drawn from a 27-category codebook defined in BOP policy documentation PS 1330.18. Discipline/Institutional Operations and Medical complaints are the two largest categories, together accounting for 30.5% of all filings. Approximately 12% of filings carry subject codes outside the standard codebook and are classified as Unknown.



Variable Descriptions:

CASENBR ***(String)***

- Unique identifier for the complaint case

sitdtrcv ***(Date)***

- Date the filing was received by the BOP. Used as the time index throughout the analysis

ITERLVL ***(Categorical)***

- Appeal level at which the filing was submitted. A = Administrative (Warden), R = Regional (Director), F = Final (Central Office)

CDFCLEVN ***(String)***

- Code identifying the facility where the complaint event occurred. 220 unique values

CDFCLRCV ***(String)***

- Code identifying the facility that received the filing

CDOFCRCV ***(String)***

- Code identifying the office that received the filing

CDSTATUS ***(Categorical)***

- Outcome of the filing. REJ = Rejected, CLD = Closed–Denied, CLO = Closed, CLG = Closed–Granted, ACC = Accepted

CDSUB1PR ***(Integer)***

- Primary subject code indicating the complaint category (e.g., 20 = Discipline, 26 = Medical). Mapped to human-readable labels using the BOP PS 1330.18 codebook

CDSUB1SC ***(String)***

- Secondary subject subcode providing additional category detail

sdtdue ***(Date)***

- Date by which a response to the filing was due

sdtstat ***(Date)***

- Date of the most recent status update on the filing

accept, reject, deny, grant, other ***(Binary)***

- Indicator flags corresponding to the filing outcome

submit, filed, closed ***(Binary)***

- Flags indicating the workflow stage of the filing

timely, untimely, overdue ***(Binary)***

- Flags indicating whether the BOP response was timely, untimely, or overdue

year_month ***(Derived)***

- Year-month period aggregated from sitdtrcv. Used as the primary time index in all analysis scripts

complaints ***(Derived)***

- Count of filings per year_month period. The univariate monthly series analyzed throughout the project

Section 2: BOP Staffing Estimates

Each row of this dataset represents one calendar year of approximate Bureau of Prisons staffing and population figures. The data were compiled manually from published BOP Annual Reports, DOJ Inspector General staffing reports, and the Congressional Research Service report R48826 (January 2026). These figures are directional approximations rather than precise administrative counts, and should be interpreted accordingly.

Variable	Mean	Min	Max
Correctional officers	17,362	14,900 (2023)	19,500 (2013)
Inmate population	185,333	145,000 (2000)	219,000 (2013)
Inmates per officer	10.7	9.1 (2000)	120 (2023)

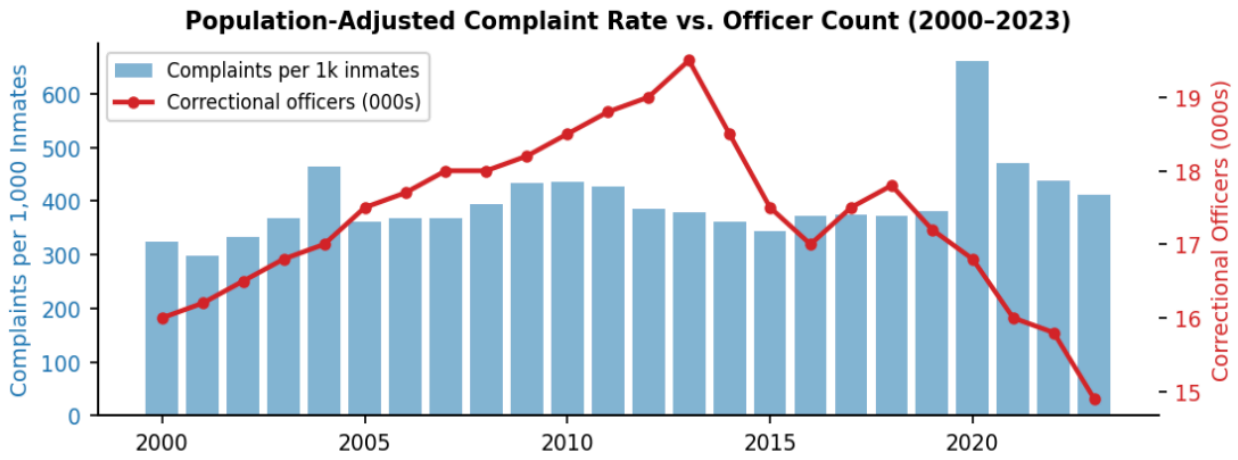


Figure 5 — Population-adjusted complaint rate (bars, left axis) vs. approximate correctional officer count (line, right axis). After population adjustment, no significant correlation between staffing levels and complaint rates is found (Pearson r = −0.24, p = 0.26).

Variable Descriptions:

year ***(Integer)***

- Calendar year. Covers 2000–2023 (24 observations)

total_staff ***(Integer)***

- Approximate total BOP staff headcount for the year, drawn from BOP Annual Reports

correctional_officers ***(Integer)***

- Approximate number of correctional officers employed by the BOP for the year. This is the primary staffing variable used in correlation analysis

inmate_population ***(Integer)***

- Approximate average annual federal inmate population, drawn from BOP Annual Reports and DOJ OIG reports

source ***(String)***

- Citation for the figures in that row (e.g., BOP Annual Report, DOJ OIG Staffing Report)

inmates_per_officer ***(Derived)***

- Inmate population divided by correctional officer count. Used as the primary staffing stress indicator in correlation analysis

complaints_per_1k_inmates ***(Derived)***

- Annual complaint count divided by inmate population, multiplied by 1,000. Removes the confounding effect of prison system size changes over time